

## Course overview

CS 4780 Applied Machine Learning is a 4-credit course taught at the Northbridge Institute of Technology. It covers supervised learning, model evaluation, neural networks and retrieval-augmented generation. Lectures take place on Mondays and Wednesdays from 10:00 to 11:30 in the Halvorsen Building, room 1.14.

## Course staff

The lecturer is Dr. Miriam Okonkwo-Lindqvist. The two teaching assistants are Tomas Beaulieu and Priya Ramanathan. The course mailbox is [cs4780-staff@northbridge.example](mailto:cs4780-staff@northbridge.example); staff answer emails within two working days.

## Office hours

Dr. Okonkwo-Lindqvist holds office hours on Tuesdays from 14:00 to 16:00 in room 3.07. Teaching assistant office hours are on Thursdays from 16:00 to 18:00 in the Ada Lab. During the final project weeks, an extra office hour runs on Fridays at 12:00 online.

## **Grading**

The final grade is made up of four parts: weekly quizzes count for 10 percent, the three lab assignments count for 30 percent, the midterm exam counts for 20 percent, and the final project counts for 40 percent. A final grade of 55 percent or higher is required to pass the course.

## **Quizzes**

Quizzes are released every Monday at 09:00 and close on Sunday at 23:59. The two lowest quiz scores are dropped automatically. Quizzes cannot be retaken.

## **Midterm exam**

The midterm exam is a 90-minute written exam held in week 8. Students may bring one double-sided A4 sheet of handwritten notes. Calculators are allowed, but phones and laptops are not.

## **Late submission policy**

Lab assignments submitted late lose 10 percent of the available points per day, for a maximum of three days. After three days the submission receives zero points. Every student has two grace days for the whole semester that can be used on lab assignments without penalty. Grace days cannot be used for the final project.

## **Extensions**

Extensions beyond the grace days are only granted for documented illness or family emergencies. Requests must be sent to the course mailbox before the deadline whenever possible, together with supporting documentation.

## **Final project**

The final project is done in teams of three students. Teams must register their topic by the end of week 6. Each team builds a working machine learning application, writes a report of at most eight pages, and gives a ten-minute live demo in week 14.

## **Project grading**

The project grade is split into the working demo (40 percent), the written report (35 percent), code quality and documentation (15 percent), and the peer review (10 percent). All team members receive the same grade unless the peer review shows a clearly unequal contribution.

## **Project rules**

Code must be kept in a version-controlled repository that the teaching assistants can access. Pretrained models and public APIs may be used, but the report must state clearly which parts the team built themselves.

## Lab schedule

Lab 1 (data cleaning and baselines) is due in week 3. Lab 2 (model evaluation and cross-validation) is due in week 5. Lab 3 (neural networks with PyTorch) is due in week 10. All labs are due on Fridays at 17:00 and are submitted through the course portal.

## Compute resources

Every student receives 40 GPU hours on the Northbridge Orca cluster. Extra hours can be requested by the team leader for the final project, up to 60 additional hours per team. Jobs that run longer than 12 hours are stopped automatically.

## **Academic integrity**

Students may discuss ideas with each other, but all submitted code and text must be their own. Using AI assistants is allowed for lab assignments only if the use is declared in a short note at the top of the submission. Undeclared use is treated as plagiarism and reported to the examination board.

## **Attendance**

Lecture attendance is not mandatory, but attendance at the week 14 project demos is required for all students. Missing your own team's demo without a valid reason results in a zero for the demo component.